

Problem 1.1

Consider the SVM dual problem:

$$\begin{aligned} \sup_{\alpha} \quad & \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j \langle x_i, x_j \rangle \\ \text{s.t.} \quad & 0 \leq \alpha_i \leq C, \forall i, \quad \sum_i \alpha_i y_i = 0. \end{aligned}$$

- Derive a coordinate ascent algorithm by fixing all the coordinates of α except two, say α_i and α_j , and optimizing only over those two, then repeating the process.
- Try to devise a heuristic for picking which pair to optimize over at each iteration.
- Implement the method and try it out on a binary classification problem where the two class are $N(\theta^{(i)}, I_2)$ with $\theta^{(1)} = (1/2, 1/2)$ and $\theta^{(2)} = (-1/2, 1/2)$. Train the SVM on $n = 100$ sample, 50 from each class. Use Monte Carlo simulation to estimate the average misclassification error and plot it as a function of C , the regularization parameter over a reasonable range of C .

Problem 1.2

Consider surrogate margin-based classification loss functions (i.e., the target loss is the 0-1 loss), $L(y, t) = \phi(yt)$. For each of the following choices:

$$\phi(t) = (1 - t)^2, \quad \phi(t) = (1 - t)_+^2, \quad \phi(t) = e^{-t}$$

answer the following questions:

- Is the loss classification calibrated?
- Find the H function.
- Find the biconjugate of the calibration function $\delta_Q^{**}(\varepsilon)$ (or $\psi^{**}(\varepsilon)$).
- Derive an inequality relating the excess risk for the surrogate loss and the excess risk for the 0-1 loss.

For the definitions of H and δ^{**} see the lecture slides.

Problem 1.3

Consider SVM with the squared hinge loss

$$\min_{w,b} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \phi(y_i(w^T x_i + b)) \tag{1}$$

where $\phi(t) = (1 - t)_+^2$. Find the dual problem.

Problem 1.4

The goal of the problem is to derive a dual for (1) when $\phi : \mathbb{R} \rightarrow \mathbb{R}$ is a general convex non-increasing function.

- (a) Argue that (1) is a convex problem (given that ϕ is convex).
- (b) Write the objective as $\frac{1}{2}\|w\|^2 + C \sum_{i=1}^n \phi(\xi_i)$ and show that with appropriate constraints, optimizing this objective over w, b and ξ_i is equivalent to (1).
- (c) Write down a dual for this problem using the conjugate function of ϕ :

$$\phi^*(u) := \sup_{t \in \mathbb{R}} [ut - \phi(t)], \quad u \in \mathbb{R}.$$

- (d) Specialize your result to $\phi(t) = e^{-t}$ and $\phi(t) = (1 - t)_+$.